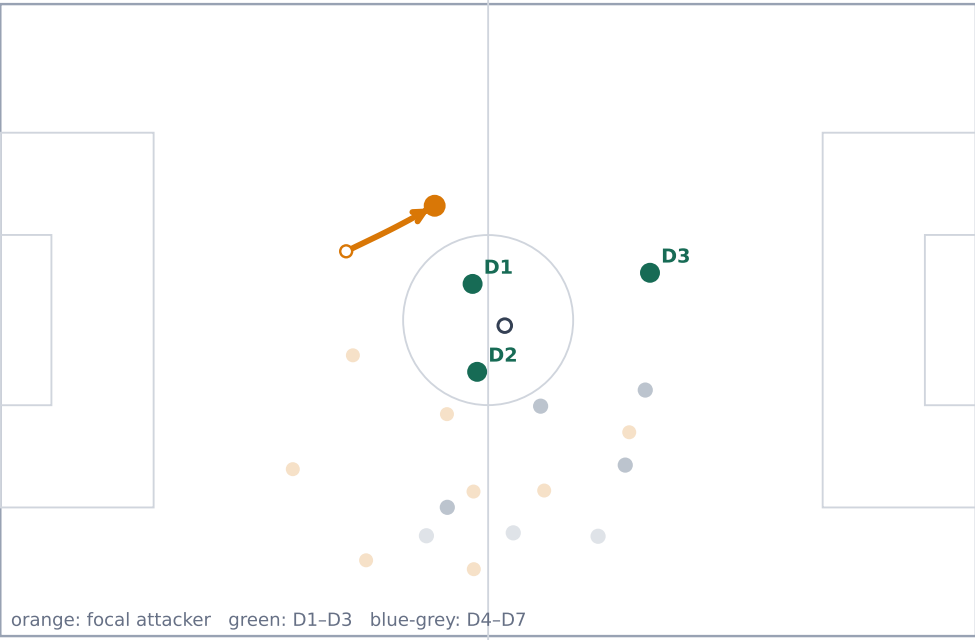


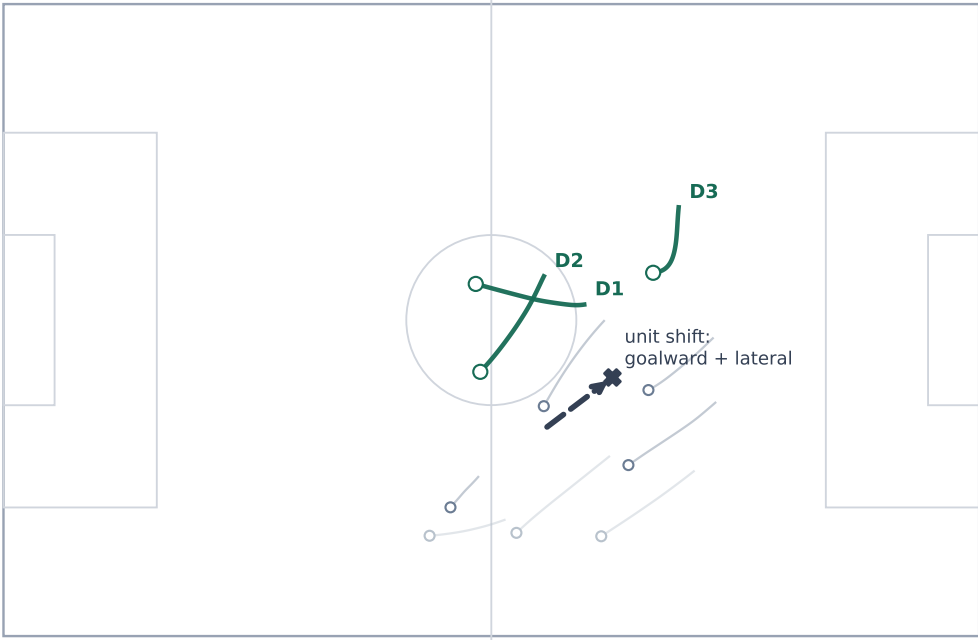
Measuring Localized Defensive Reorganization Associated with Off-Ball Movement in Football

Panel A: deterministic heldout Metrica Game 2 anchor at 2336.04 s (earliest eligible anchor at or above the upper quartile of preceding attacker path; selection uses attacker movement only).

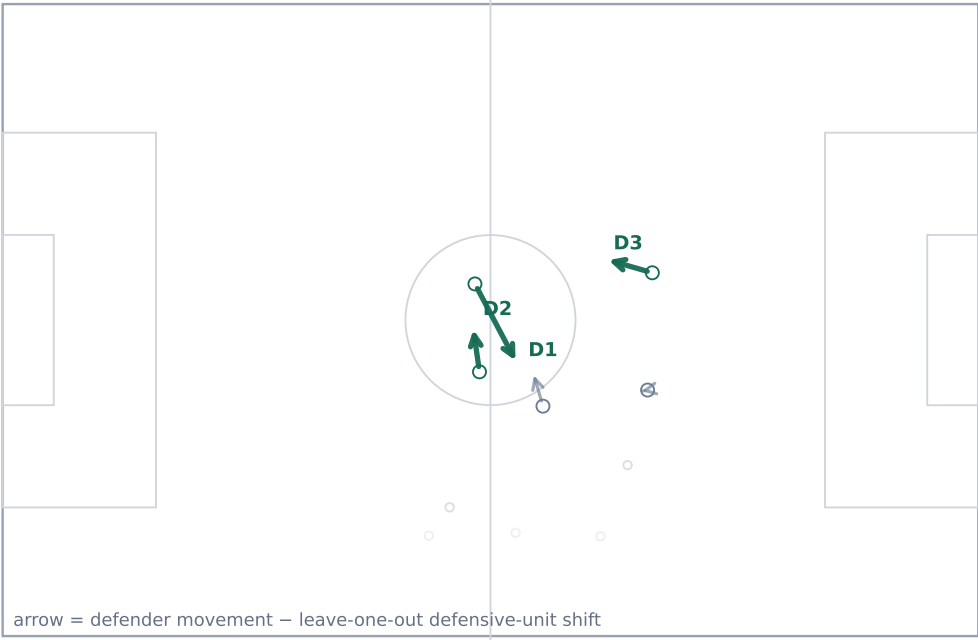
A1 Exposure: preceding attacker movement



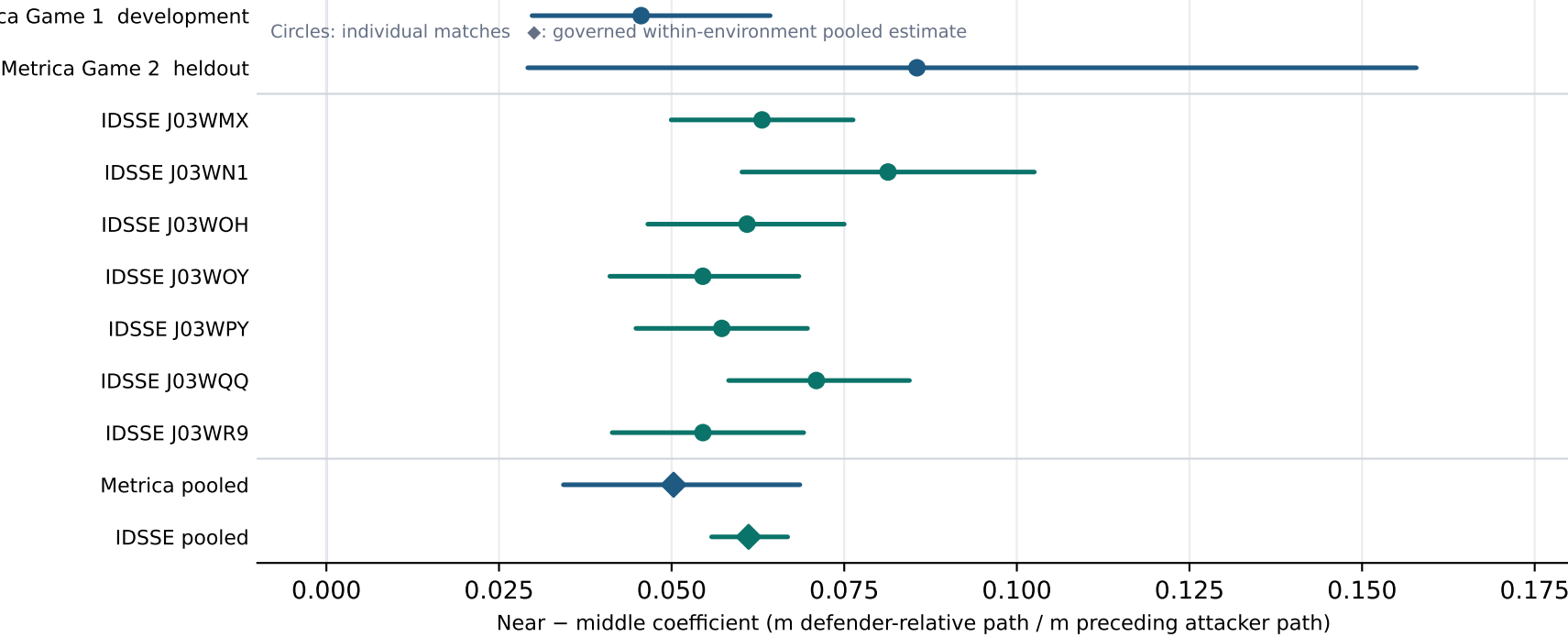
A2 Next 2 s: absolute defender movement



A3 Same movement relative to the defensive unit

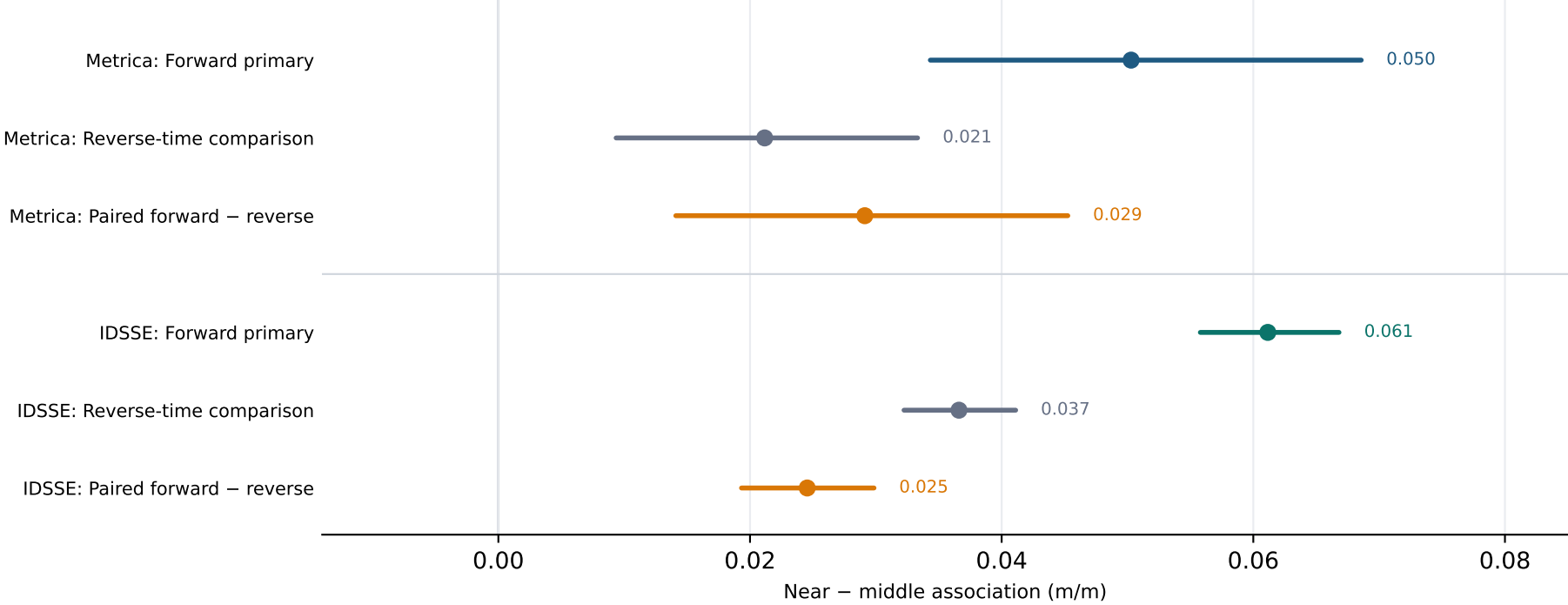


B The localized association replicates across matches



Replicated, localized, time-ordered defensive reorganization. Not established: causation, tactical meaning, opportunity, or value.

C Forward association exceeds reverse time



Reverse-time structure remains positive. The evidence is paired forward – reverse excess, not a reverse-time null.